

Quantum Key Distribution

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Agenda

Introduction

Fundamental Principles about Quantum Key Distribution

Protocols, Security, and Quantum Bit Errors

Practical Experiences from Conducted Projects

Summary

LECTURE OBJECTIVE

Explore Quantum Key Distribution (QKD)

as a promising solution

to

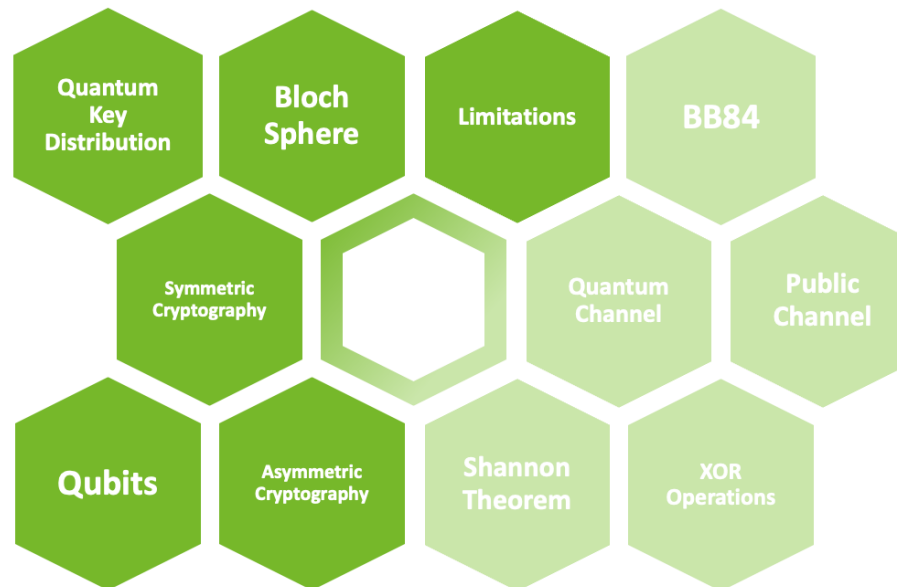
distribute cryptographic key material in a secure manner

adhering to the principles of quantum mechanics and

using those keys to encrypt sensitive data.

Nice – But why is that?

KEYWORDS AT ONE VIEW



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CLASSIC CRYPTOGRAPHY

SELECTED EXAMPLES

Ubiquitous on the internet

- IPsec¹/IPv6² (*or alternatively IPv4*)
- SSL³/TLS⁴ (*e.g., https⁵, IMAPS⁶, POPS⁷*) or DTLS⁸
- SSH⁹
- Kerberos
- X.509 Certificates

¹ IPsec – Internet Protocol Security (IP Security)

² IPv6 – Internet Protocol Version 6

³ SSL – Secure Sockets Layer

⁴ TLS – Transport Layer Security

⁵ https – Hypertext Transfer Protocol Secure

⁶ IMAPS – Internet Message Access Protocol

⁷ POPS – Post Office Protocol Secure

⁸ DTLS – Datagram Transport Layer Security

⁹ SSH – Secure Shell

CLASSIC CRYPTOGRAPHY

CENTRAL CONCEPTS

2 Central Approaches

- Public key cryptography (e.g., *Diffie Hellman*, *ECC*¹, *ECDSA*², *RSA*³)
- Symmetric key cryptography (e.g., ***one time pad***, *AES*⁴, *DES*⁵, *TDES*⁶)

The Security is Mainly Based on Complexity Theory

- Security often relies on unproven assumptions concerning the complexity of mathematical problems
- RSA: complexity of integer factorization?
- Diffie Hellman, ECC: complexity of discrete logarithm?

¹ ECC – Elliptic Curve Cryptography

² ECDSA – Elliptic Curve Digital Signature Algorithm

³ RSA – Rivest Shamir Adleman

⁴ AES – Advanced Encryption Standard

⁵ DES – Data Encryption Standard

⁶ TDES – Triple DES

PROTECTION AGAINST QUANTUM COMPUTERS

SUGGESTED SOLUTION

Protect assets with Quantum Key Distribution and achieve private communication based on the laws of physics

- Security is based on the properties of quantum physics
- Eavesdropper detection
- Randomness derived from quantum mechanics
- Unconditional security by contrast to complexity assumptions
- Not a „*cryptography technique*“ such as AES, RSA

Quantum Key Distribution is a Promising Method to Distribute Cryptographic Key Material in a Secure Manner

WHAT ARE NOW 4 RELEVANT PRINCIPLES IN QKD? COMBINING PHYSICS AND INFORMATION THEORY

4 relevant principles in terms of Quantum Cryptography are:

- (1) The term „*Quantum Cryptography*“ is a bit imprecise¹ – „*Quantum Key Distribution*“ (QKD) is more accurate
- (2) Quantum Keys are distributed adhering to the principles of Quantum Mechanics and enable Eavesdropper detection (*based on the principles of quantum physics*)
- (3) For encryption of (e.g., sensitive) data **conventional methods** are applied (*such as the one-time pad*)
- (4) Quantum Key Distribution is pure elegance and basically a relative simple idea

WHAT ARE 5 COUNTERINTUITIVE STATEMENTS IN QM?

QM is well-known for being counterintuitive, or even bizarre: Five ‘negative’ rules to be derived

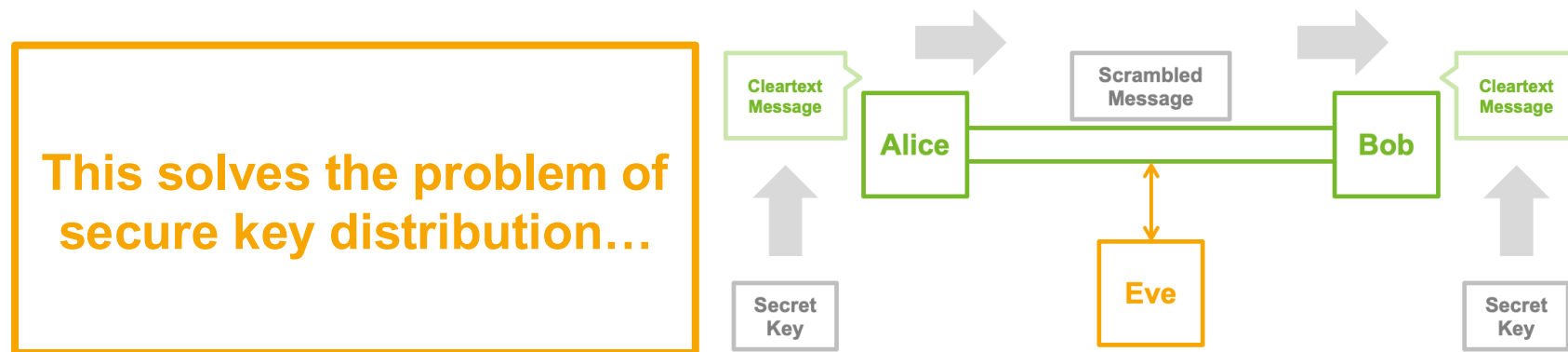
- Every measurement perturbs the system (*very important*)
- One cannot determine simultaneously the position and the momentum of a particle with arbitrary high accuracy (*“Heisenberg’s uncertainty relation”*)
- One cannot measure the polarization of a photon in the vertical-horizontal basis and simultaneously in the diagonal basis
- One cannot draw pictures of individual quantum processes¹
- One cannot duplicate an unknown quantum state (*the famous “no cloning theorem”*)

THE BASIC COMMUNICATIONS MODEL

QUANTUM KEY DISTRIBUTION SIMPLIFIED

Alice and Bob with (*malicious*) Eavesdropper „Eve“

- Authenticated channel and public (*e.g., unprotected internet*) channel
- Bob is able to verify that a message came from Alice
- Alice is able to verify that a message came from Bob



SYMMETRIC CRYPTOSYSTEMS

ENCRYPTION AND DECRYPTION

- Symmetric ciphers require the use of a single key for both encryption and decryption.
- The “one-time pad” (or: *Vernam cipher*, 1926) belongs to this category
- A encrypts her message, a string of bits denoted by the binary number m_1 , using a randomly generated key k .
- The scrambled text is s and can be encoded using k by applying XOR and decoded again by applying XOR using the same key k

$$s = m_1 \oplus k \longrightarrow s \oplus k = m_1 \oplus k \oplus k = m_1$$

- This cryptosystem is provably secure in the sense of information theory (*Shannon*, 1949) assumed k is randomly generated
- Although perfectly secure, A and B have to possess a common secret key, which must be at least as long as the message itself
- A and B can only use the key for a single encryption (hence “one-time pad”)

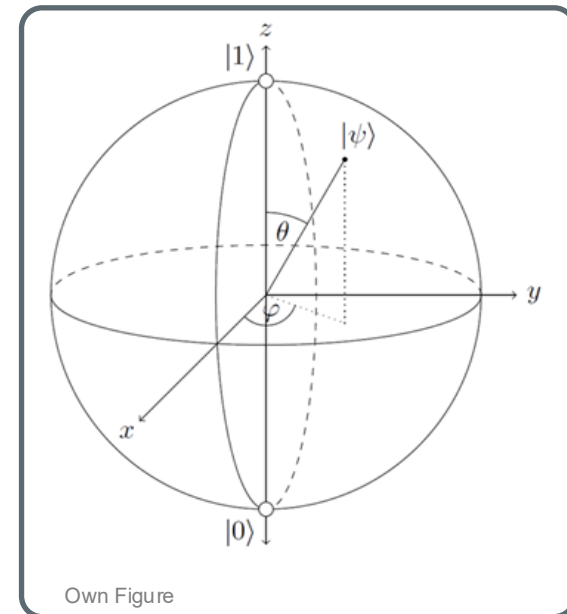
WHAT ARE NOW INFORMATION CARRIERS IN QKD?

BITS VS. QUBITS

Representation of Quantum Systems

- Bloch sphere
- Introduced by Felix Bloch (23.10.1905 – 10.09.1983)
- Geometrical representation
- Quantum mechanical system
- $0 \leq \theta \leq \pi, 0 \leq \varphi \leq 2\pi$

Information carriers are „qubits“



WHO ARE ALICE AND BOB IN QKD? SENDER AND RECEIVER

QM is well-known for being counterintuitive, or even a portion of „bizarre“: Five ‘negative’ rules

- ... except the quantum system is compatible with the measurement !
- The positive side of this rule 1: application to a communication between Alice (A), and Bob (B), *(the conventional names of sender – A – and receiver – B, respectively)*
- The support of information are quantum systems
- For example, individual photons
- Rule 1 applies also to the eavesdroppers, i.e. to a malicious Eve, E, *(the conventional name given to the adversary in cryptology)*
- Hence, E cannot get any information about the communication without introducing perturbations which would reveal her presence
- More precise: A directly encodes information in individual photons and sends them to B

WHO ARE ALICE AND BOB IN QKD? | cont'd

SENDER AND RECEIVER

- B receives the photons unperturbed, then, by the basic rule 1, the photons were not measured
- No measurement implies: E did not get any information
- Consequently: A and B can check whether someone was listening
- They simply compare a randomly chosen subset of their data using a public channel
- A secret quantum key can be derived which is distributed between A and B to encode and decode messages

No perturbation means no measurement and no eavesdropping

WHAT DOES THAT MEAN?

2 additional aspects to be taken into account

- a) A encodes her information in non-orthogonal states (*see next chapter*).
- b) A and B do not use the quantum channel to transmit information, but only to transmit a random sequence of Bits, i.e. a key.
 - If the key is unperturbed, then QM guarantees that no one got any information about this key by eavesdropping (*i.e. by measuring*) the quantum communication channel.
 - If, on the contrary, the key turns out to be perturbed, then A and B simply disregard it.

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POLARIZATION

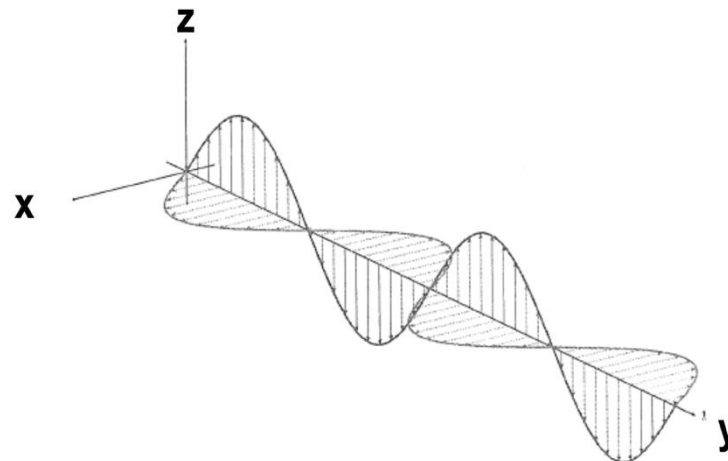
BACKGROUND INFORMATION

The direction of the electric field of an electromagnetic field is called the polarization.

Properties

- 1) Linear Polarization
- 2) Circular Polarization
- 3) Elliptical Polarization
- 4) Unpolarized Light
- 5) Birefringence

Horizontal (H) and vertical polarized (V) light



POLARIZATION | cont'd

BACKGROUND INFORMATION

- 1) **Linear polarization:** the electrical field vector points along a constant direction.
45° linear polarization is proportional to (H + V)
- 2) **Circular polarization:** the electrical field vector rotates as the wave propagates. Right circularly polarized Left circularly polarized
Circularly polarized light can be decomposed into 2 orthogonal linearly polarized waves of equal amplitude with a 90° phase difference between them
- 3) **Elliptical polarization:** Similar to circular polarization, except different amplitudes.
- 4) **Unpolarized light:** the light is randomly polarized.

POLARIZATION | cont'd

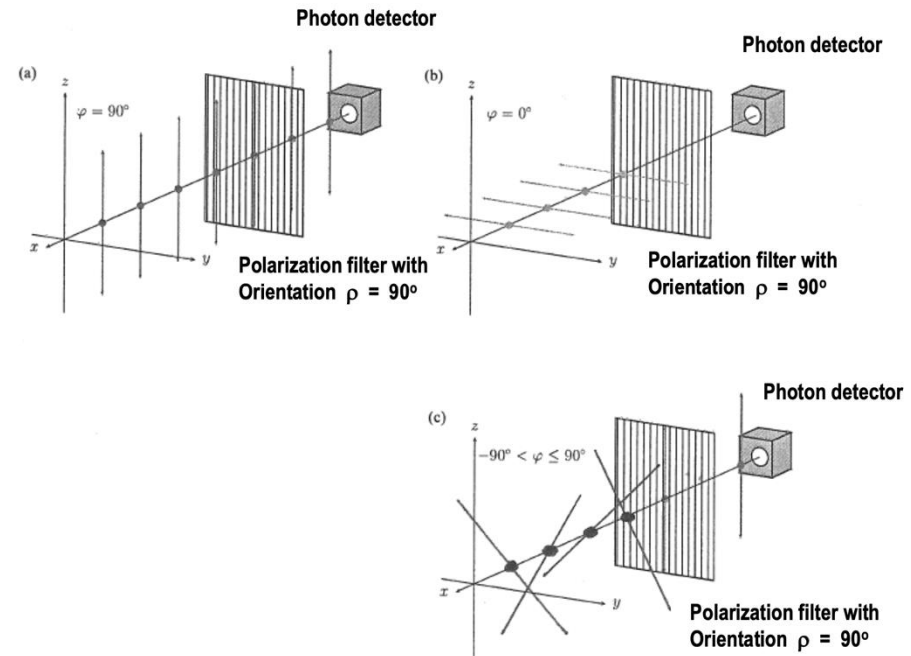
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Circularly polarized light can be decomposed into 2 orthogonal linearly polarized waves of equal amplitude with a 90° phase difference between them
- 3) **Elliptical polarization:** Similar to circular polarization, except different amplitudes.
- 4) **Unpolarized light:** the light is randomly polarized.
- 5) **Birefringence:** In free space the polarization of a wave is constant as it propagates. In certain anisotropic materials, the polarization can change as the wave propagates. Birefringent crystals separate arbitrarily polarized beams into 2 orthogonally polarized beams called the ordinary ray and the extra-ordinary ray (*different refractive indices!*).
- 6) **Polarizing beam splitter (PBS):** commonly made by cementing together two birefringent materials like calcite or quartz, and has the property of splitting a light beam into its orthogonal linear polarizations.

GENERAL OVERVIEW MEASUREMENT SETUP

General Setup

- Polarized photons transmitted
- Polarized photons impact polarization filter
- Polarization Filter “decides“
- Detector is able to detect/click



EXAMPLES FROM TRANSMISSIONS

Input ψ	Measurement ρ	Detector D	Probability	V(x)
↑	↑	↑	100%	0
↑	↗	↗	50%	1
→	↑		0%	0
→	↗	↗	50%	1

EXAMPLES FROM TRANSMISSIONS

Input ψ	Measurement ρ	Detector D	Probability	V(x)
			50%	1
			0%	0
			50%	1
			50%	1

EXAMPLES FROM TRANSMISSIONS

Input ψ	Measurement ρ	Detector D	Probability	V(x)
$\uparrow$	$\times$	$\times$	50%	1
$\nearrow$	$+$	$+$	50%	1
$\rightarrow$	$\times$	$\times$	50%	1
$\rightarrow$	$+$	$\rightarrow$	100%	0

EXAMPLES FROM TRANSMISSIONS

Input ψ	Measurment ρ	Detector D	Probability	V(x)
			100%	0
			50%	1
			100%	0

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BB84

AN EXAMPLE

- The BB84 protocol introduced by Charles H. Bennett and Gilles Brassard
- 100% perfect security exists

The necessities are

- Random key material
- Distributed between communication partners (Alice and Bob)
- Used only once
- Being the same length as the message

Result: One Time Pad (or Vernam Scheme/Chiffre)

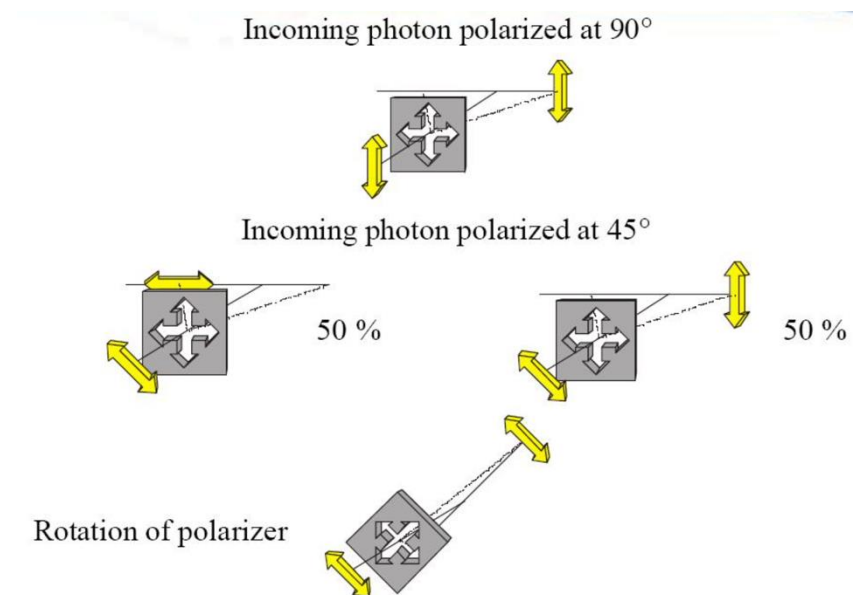
BB84 | cont'd

AN EXAMPLE

Which problems arise: security of key distribution and absolute randomness

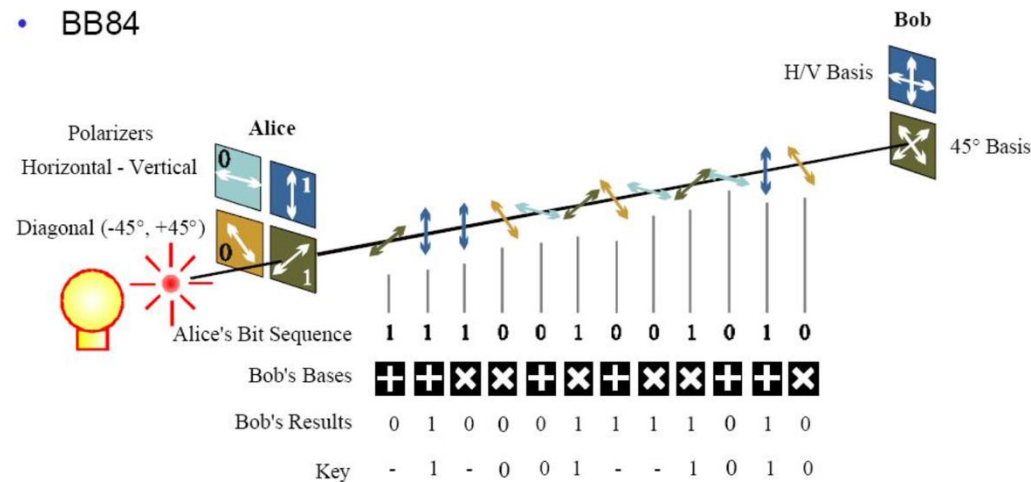
- The laws of quantum mechanics guarantee absolute randomness of bit sequences and security against eavesdropping (secure key distribution)
- Method to go: quantum key distribution, i.e. the distribution of a quantum property of single photons (polarization) via a quantum channel (fibre glass or free space)

BB84 | IRREVERSIBILITY OF A MEASUREMENT



BB84 EXAMPLE TRANSMISSIONS OF QUBITS

- BB84



Better than „Quantum Cryptography“ -> „Quantum Key Distribution“

4 PROCESSING STEPS IN BB84

- (1) Sending (Alice) polarized photons on the quantum channel and receiving those (Bob)
- (2) Public Channel interaction to cleanup the measured photons
- (3) Public Channel interaction to derive the raw key (i.e., base comparison)
- (4) Public Channel error estimation and error correction

4 PROCESSING STEPS IN BB84 | cont'd

Sending (Alice) polarized photons on the quantum channel and receiving those (Bob)

Alice (A) sends randomly polarized photons across the quantum channel to Bob (B)

The 4 basic vectors are sent: $|H\rangle$, $|V\rangle$, $|+\rangle$ (or $|R\rangle$), $|-\rangle$ (or $|L\rangle$)

Corresponding to the polarization directions $\varphi = 0^\circ, 90^\circ, +45^\circ, -45^\circ$

2 bases are generated

- a) The rectilinear (or rectangular) basis with vectors $|H\rangle$, $|V\rangle$
- b) The diagonal (or rotated) basis with vectors $|+\rangle$, $|-\rangle$

Bit 1 ... $|H\rangle$ or $|+\rangle$ vector

Bit 0 ... $|V\rangle$ or $|-\rangle$ vector

Bob (B) receives the photons

4 PROCESSING STEPS IN BB84 | cont'd

Sending (Alice) polarized photons on the quantum channel and receiving those (Bob)

- Bob uses 4 polarization filters with the same orientations $\rho = \{0^\circ, 90^\circ, +45^\circ, -45^\circ\}$
- The orientations are distributed on a random basis
- The orientations and results of (the synchronized) measurements are quoted by A and B
- Line 2. A decided randomly to keep basis (+) and a vertical polarized photon $|V\rangle$ (bit 0). B takes randomly and coincidentally the same basis. However, using a horizontal filter orientation $\rho = 0^\circ$. He therefore does not register a photon (no bit (X), no detection).
- Line 9. A has again the same basis (+) and a vertically polarized photon (bit 0). B has now a filter orientation $\rho = 90^\circ$ and he registers a photon (bit 0).
- Line 11. A and B have different bases. The probability for a horizontally polarized photon to pass Bobs filter $\rho = +45^\circ$ is exactly $\frac{1}{2}$. Those items are treated in the next step – i.e., step 3.

BB84 SUMMARIZED

STEP 1

Alice				Step 1	Bob			
Nr.	Basis	Photon ψ	Bit	Send and measure	Basis	Filter ρ	Detection	Bit
1	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
2	$\oplus$	$ V\rangle$	0		$\oplus$	0°	○	×
3	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
4	$\otimes$	$ -\rangle$	0		$\oplus$	0°	•	1
5	$\oplus$	$ H\rangle$	1		$\otimes$	-45°	•	0
6	$\otimes$	$ +\rangle$	1		$\oplus$	0°	○	×
7	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
8	$\otimes$	$ +\rangle$	1		$\otimes$	-45°	○	×
9	$\oplus$	$ V\rangle$	0		$\oplus$	90°	•	0
10	$\oplus$	$ V\rangle$	0		$\oplus$	90°	•	0
11	$\oplus$	$ H\rangle$	1		$\otimes$	$+45^\circ$	•	1
12	$\otimes$	$ -\rangle$	0		$\otimes$	$+45^\circ$	○	×
13	$\otimes$	$ -\rangle$	0		$\otimes$	-45°	•	0
14	$\oplus$	$ V\rangle$	0		$\otimes$	-45°	•	1
15	$\otimes$	$ -\rangle$	0		$\oplus$	0°	○	×
16	$\otimes$	$ +\rangle$	1		$\oplus$	$+45^\circ$	•	1
17	$\otimes$	$ +\rangle$	1		$\oplus$	90°	•	1
18	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
19	$\otimes$	$ -\rangle$	0		$\otimes$	$+45^\circ$	○	×
20	$\oplus$	$ H\rangle$	1		$\oplus$	90°	○	×
21	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
22	$\oplus$	$ V\rangle$	0		$\otimes$	$+45^\circ$	•	1
23	$\otimes$	$ -\rangle$	0		$\otimes$	-45°	•	0
24	$\otimes$	$ -\rangle$	0		$\oplus$	0°	○	×

BB84 SUMMARIZED | cont'd

Public Channel interaction to cleanup the measured photons

- From now on all submitted information is delivered via the public channel (i.e., conventionally such as unsecure internet, phone, e-mail, or even with fax)
- B informs A about registered photons
- Both parties (i.e., A and B) erase entries where not photon was registered by B
- Consequently, entries 2, 6, 8, 12, 15, 19, 20, 24 are eliminated
- It should be mentioned that an incoming photon not always causes an appropriate „detector click“. The detector efficiency plays an important role.
- Now we have eliminated entries where A and B have the same basis as well. For instance, B could have chosen a filter rotated by 90° in comparison to the filter of A (cf., entry 2 above). Those entries are useful and secure as well – to be highlighted and should not be discarded. Below (see “measuring device for Bob”) we shall become acquainted with a measuring method where this can be taken into account.
- Using this measuring method step 2 can be omitted in principle.

BB84 SUMMARIZED | cont'd

STEP 2

Alice				Step 2	Bob			
Nr.	Basis	Photon ψ	Bit	Filtering of detected photons	Basis	Filter ρ	Detection	Bit
1	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
2	$\oplus$	$ V\rangle$	0		$\oplus$	0°	○	×
3	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
4	$\otimes$	$ -\rangle$	0		$\oplus$	0°	•	1
5	$\oplus$	$ H\rangle$	1		$\otimes$	-45°	•	0
6	$\otimes$	$ +\rangle$	1		$\oplus$	0°	○	×
7	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
8	$\otimes$	$ +\rangle$	1		$\otimes$	-45°	○	×
9	$\oplus$	$ V\rangle$	0		$\oplus$	90°	•	0
10	$\oplus$	$ V\rangle$	0		$\oplus$	90°	•	0
11	$\oplus$	$ H\rangle$	1		$\otimes$	$+45^\circ$	•	1
12	$\otimes$	$ -\rangle$	0		$\otimes$	$+45^\circ$	○	×
13	$\otimes$	$ -\rangle$	0		$\otimes$	-45°	•	0
14	$\oplus$	$ V\rangle$	0		$\otimes$	-45°	•	1
15	$\otimes$	$ -\rangle$	0		$\oplus$	0°	○	×
16	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
17	$\otimes$	$ +\rangle$	1		$\oplus$	90°	•	1
18	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
19	$\otimes$	$ -\rangle$	0		$\otimes$	$+45^\circ$	○	×
20	$\oplus$	$ H\rangle$	1		$\oplus$	90°	○	×
21	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
22	$\oplus$	$ V\rangle$	0		$\otimes$	$+45^\circ$	•	1
23	$\otimes$	$ -\rangle$	0		$\otimes$	-45°	•	0
24	$\otimes$	$ -\rangle$	0		$\oplus$	0°	○	×

BB84 SUMMARIZED | cont'd

Public Channel interaction to derive the raw key (i.e., base comparison)

- (1) A and B compare their bases using the public channel (e.g., unsecure internet, phone, or fax)
- (2) For different bases (regarding the same entry) the corresponding Bits can only be equal accidentally
- (3) Therefore, if the bases are different, they discard the data of such entries (4, 5, 11, 14, 17, 22)
- (4) Remark: A and B must not disclose their filter positions, they only announce their bases (+) or (X) and discard entries with different bases. This is an essential aspect in terms of (unconditional) security.
- (5) In the case of equal bases from A and B should have an identical sequence of bits which is the so-called "raw key". 11 1000 1110 i.e., entries 1, 3, 7, 9, 10, 13, 16, 18, 21, 23

BB84 SUMMARIZED | cont'd

STEP 3

Alice				Step 3	Bob			
Nr.	Basis	Photon ψ	Bit	Comparison of bases	Basis	Filter ρ	Detection	Bit
1	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
2								
3	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
4	$\otimes$	$ -\rangle$	0		$\oplus$	0°	•	1
5	$\oplus$	$ H\rangle$	1		$\otimes$	-45°	•	0
6								
7	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
8								
9	$\oplus$	$ V\rangle$	0		$\oplus$	90°	•	0
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12								
13	$\otimes$	$ -\rangle$	0		$\otimes$	-45°	•	0
14	$\oplus$	$ V\rangle$	0		$\otimes$	-45°	•	1
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16	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
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19								
20								
21	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
22	$\oplus$	$ V\rangle$	0		$\otimes$	$+45^\circ$	•	1
23	$\otimes$	$ -\rangle$	0		$\otimes$	-45°	•	0
24								

BB84 SUMMARIZED | cont'd

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- (5) In the case of equal bases from A and B should have an identical sequence of bits which is the so-called "raw key". 11 1000 1110 i.e., entries 1, 3, 7, 9, 10, 13, 16, 18, 21, 23

ABSORPTION ON THE QUANTUM CHANNEL SIMULATION

Transmission losses on noisy optical fibre channels

$$t_L = 10^{-\frac{\alpha \cdot d}{10}}$$

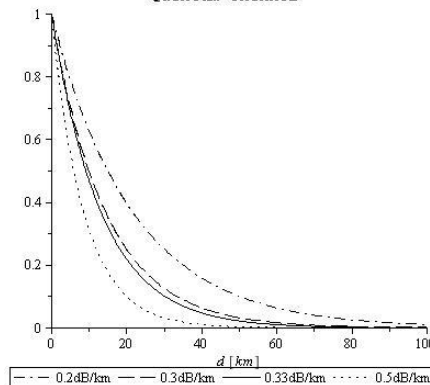
α ... absorption on the
quantum channel in dB/km

d .. distance in m

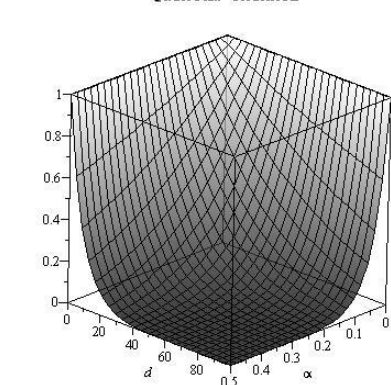
λ ... 1.500 nm

- Left: fixed α
- Right: α and d variable with
 $\alpha \in \{0.2, 0.3, 0.33, 0.5\}$ dB/km

Absorption [dB] over distance [km] on the Quantum Channel



Absorption [dB] over distance [km] on the Quantum Channel



BB84 ERROR CORRECTION AND ESTIMATION

Public Channel error estimation and error correction

- (1) The remaining Bit sequence (“raw key”) 1110001110 cannot yet be used as a key
- (2) The key has to be checked for errors (algorithms for error correction)
- (3) Observation: Eve could have eavesdropped by wiretapping
- (4) A (small) subset of the obtained “raw key” is used in order to identify the error rate. Here the first 3 Bits are chosen
- (5) If the error rate is too high (see below about the “security of the protocol”) the whole “raw key” should be discarded
- (6) If the error rate is below a certain threshold, “classical error correction” and “privacy amplification” has to be executed in order to get an error-free final key. We don’t go into the details
- (7) If the error rate is below a certain threshold, “classical error correction” and “privacy amplification” has to be executed in order to get an error-free final key. We don’t go into the details

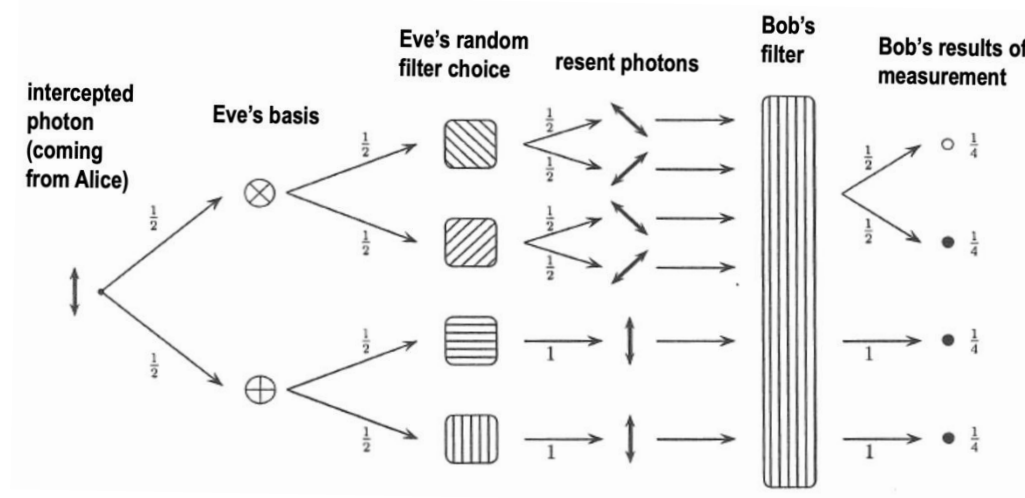
BB84 SUMMARIZED | cont'd

STEP 3

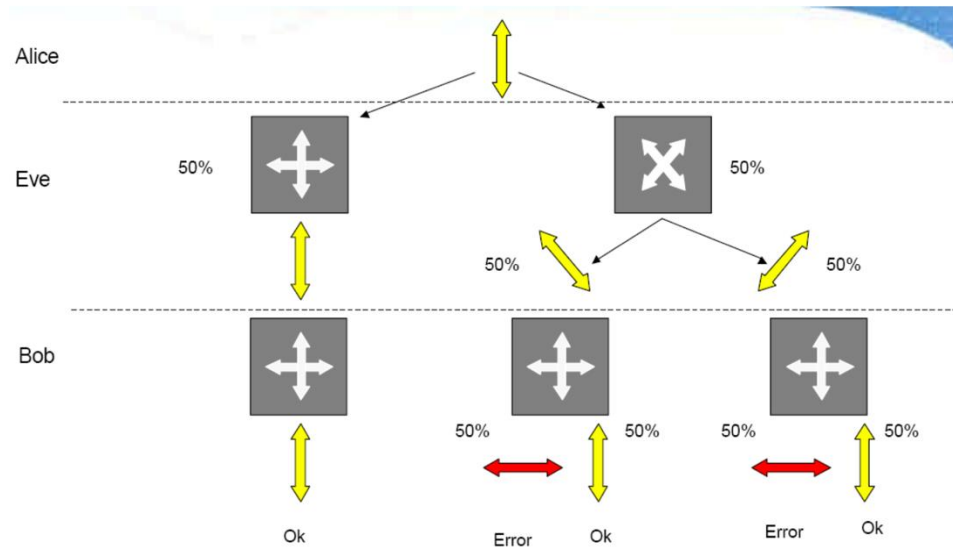
Nr.	Alice			Step 4 Comparison of errors	Bob			
	Basis	Photon ψ	Bit		Basis	Filter ρ	Detection	Bit
1	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
2								
3	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
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5								
6								
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11								
12								
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18	$\oplus$	$ H\rangle$	1		$\oplus$	0°	•	1
19								
20								
21	$\otimes$	$ +\rangle$	1		$\otimes$	$+45^\circ$	•	1
22								
23	$\otimes$	$ -\rangle$	0		$\otimes$	-45°	•	0
24								

THE SECURITY OF BB84

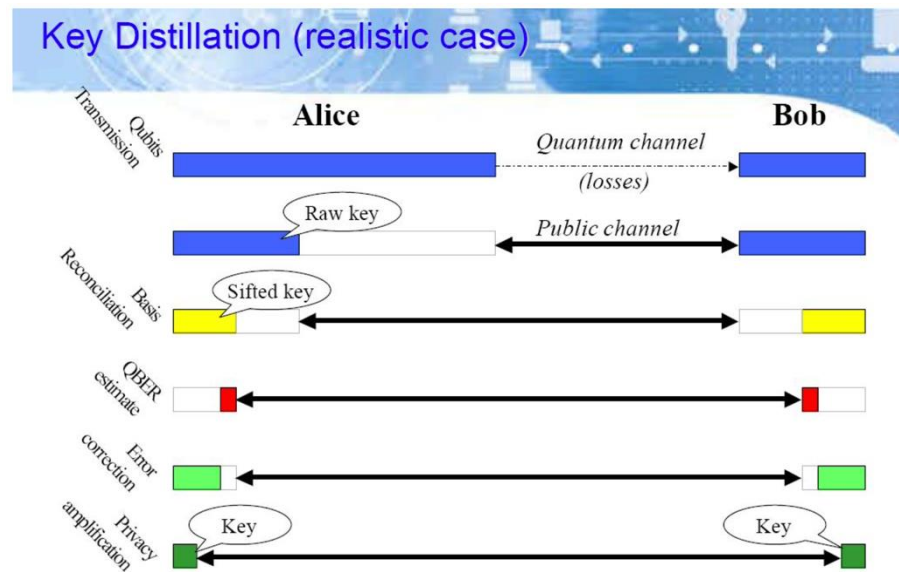
- If E measures in the (+) basis she does not change the polarization state, fortunately.
- Applying the (x) basis, E inevitably converts the state to a rotated one. This happens with 50% probability. Resending the photon to B, Bob definitely discovers such an eavesdropping attack.
- 3 almost equal examples follow



THE SECURITY OF BB84



KEY DISTILLATION OVERVIEW



Agenda

Introduction

Fundamental Principles about Quantum Key Distribution

Protocols, Security, and Quantum Bit Errors

Practical Experiences from Conducted Projects

Summary

OVERVIEW

- Noisy quantum channels +
- Transmission losses on noisy optical fibres occur
- Imperfections of e.g., measurement devices
- QKD Devices are expensive
- Unconditional security based on quantum mechanics
- Combined quantum key generation/distribution with conventional one-time-pad
- Conventional one-time-pad enables information theoretic secure encryption/decryption

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